Released trees leak training records, and the attacker knows which ones

Reconstructing training data from a published Rashomon set of decision trees

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Why it matters

- Interpretable decision trees get published for transparency; a **Rashomon set** holds all near-optimal trees [4].
- Each tree reveals how many training rows fall in every leaf; together, the counts pin down the training table [1–3].
- Earlier attacks report one error for the whole table, so the attacker cannot tell *which* rebuilt rows are real people.

Research question

When m trees of a Rashomon set are released, how many training records can an attacker rebuild, and can it tell which ones are right?

1 How do m and background knowledge affect the error?

2 Can each record get a confidence that is calibrated without ground truth?

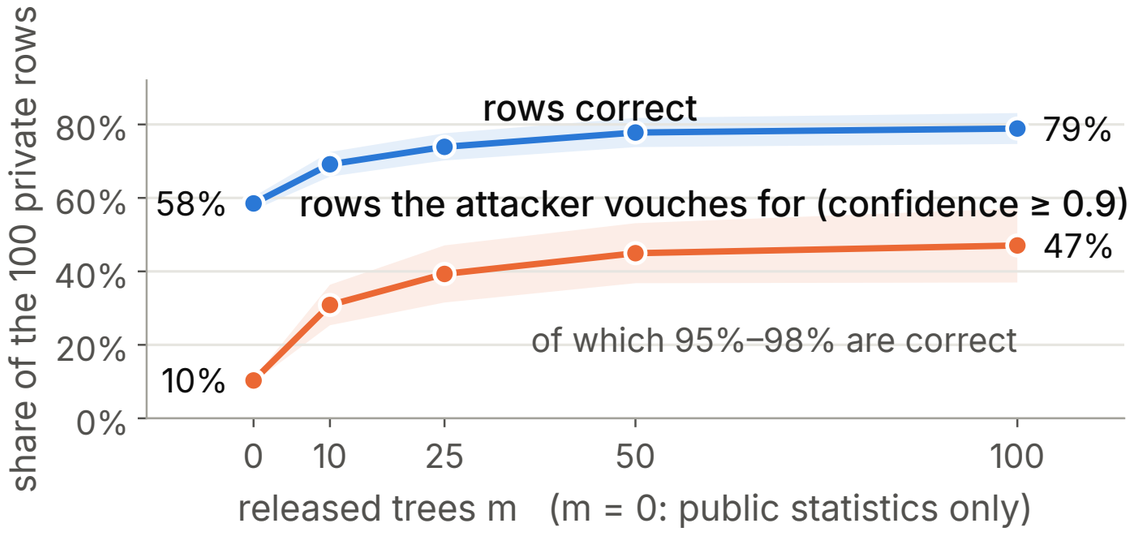
Method

Private data — 100 rows of UCI Adult, 13 binary columns + income → **Rashomon set** — all trees within 10% of the best (TreeFARMS) → **Release** — m trees drawn at random, with their leaf counts → **Attack (CP-SAT)** — 20 tables that fit every leaf count, guided by 2,000 public records → **Output** — 100-row table; a row's confidence = share of the 20 tables with it

Prior-guided sampling: each table is the consistent table closest to a random draw from the public records. **Baseline:** public statistics only, no trees. **RQ1:** also knows columns A, B; m = 1–200, 100 seeds. **RQ2:** m = 10–100, 20 seeds.

Result 1 · The trees leak records

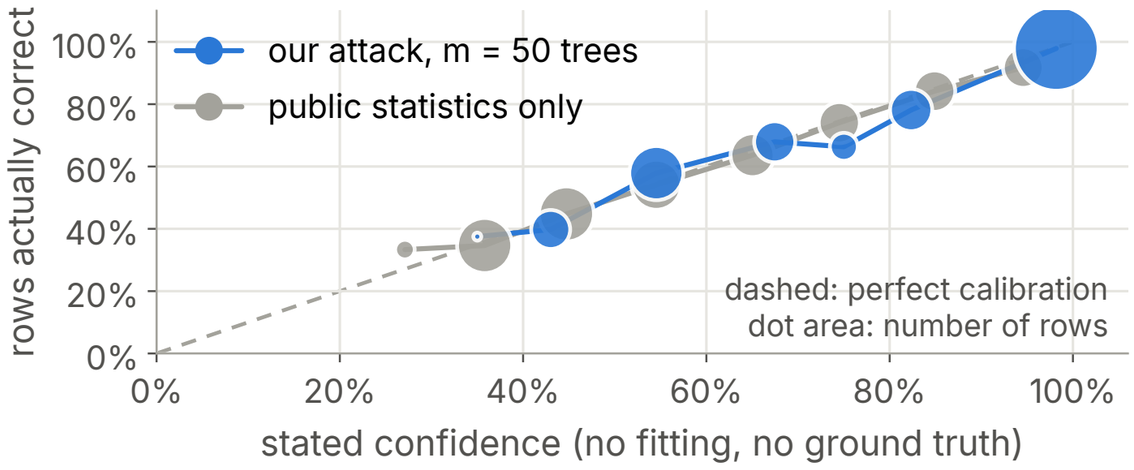
With 50 trees the attacker vouches for 45% of the rows, and 98% of those are right.



20 seeds (18 at m = 100), 95% intervals; correct = every column matches.

Result 2 · The attacker knows which

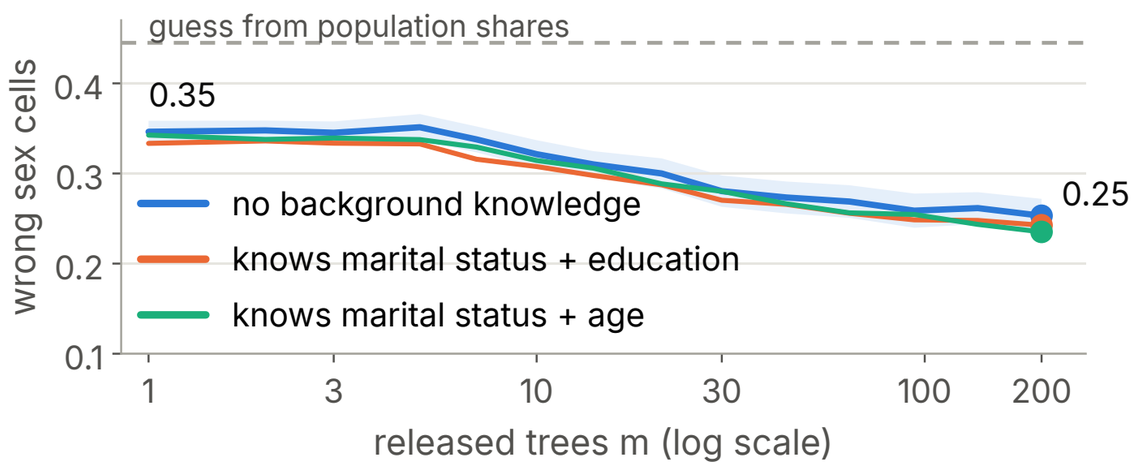
Stated confidence matches accuracy with no fitting: off by only 0.7–2.3 points at every m.



All 2,000 rows of 20 seeds at m = 50, grouped by stated confidence.

Result 3 · Knowing more barely helps

200 trees cut the error on sex from 0.35 to 0.25; two known attributes save about 1 point more.



A = marital status, B = education or age; 100 seeds; band: 95% interval.

Result 4 · What changes the answer

- **Which trees:** TreeFARMS lists near-copies first; releasing the first m trees understates the leak by 8–12 points.
- **Scarce public data:** with 200 public records the attack is 6–9 points overconfident; a Dirichlet prior brings this to 2, at equal accuracy.

Conclusion and future work

- More released trees leak more records, and the attacker knows which.
- Judge privacy risk per record, not by one table-wide error.
- **Next:** differentially private leaf counts, larger tables, public data from another population.

References and code

[1] J. Ferry, U. Aïvodji, S. Gambs, M.-J. Huguet, M. Siala. Probabilistic dataset reconstruction from interpretable models. *IEEE SaTML*, 2024.

[2] J. Ferry, R. Fukasawa, T. Pascal, T. Vidal. Trained random forests completely reveal your dataset. *ICML*, 2024.

[3] E. Hsu, H. Chen, C. Zhong, L. Semenova. The double-edged nature of the Rashomon set for trustworthy machine learning. *ICML*, 2026.

[4] R. Xin, C. Zhong, Z. Chen, T. Takagi, M. Seltzer, C. Rudin. Exploring the whole Rashomon set of sparse decision trees. *NeurIPS*, 2022.

[5] T. Dick, C. Dwork, M. Kearns, T. Liu, A. Roth, G. Vietri, Z. S. Wu. Confidence-ranked reconstruction of census microdata from published statistics. *PNAS*, 2023.

Code, data and results

github.com/francescofuligni/rtds-project